

Statistics
367-1-4361
Model Comparison
Opher Donchin

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10.1 Overfitting and underfitting

A first criterion for choosing between models

Often many models are possible

- Choosing predictors
 - Continuous and categorical
- Interactions
- Hierarchical structure
- Heteroscedasticity
- Transformations
- Likelihood choices
 - Robust likelihoods
 - Bounded likelihoods

So far we have only one type of comparison

- Null hypothesis testing
 - Reject specific models
 - Because they are not probable a posteriori
 - Because they are not likely to produce our data
 - Or accept a null model
 - If HDI is in ROPE
- We want to compare models
 - Not just reject or accept them

R^2 measures the amount of fit

- How much of the total variance is explained by the model
 - Often called 'goodness of fit'
- Always goes up with the number of parameters
 - Is the model with the highest 'goodness of fit' the best model?

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

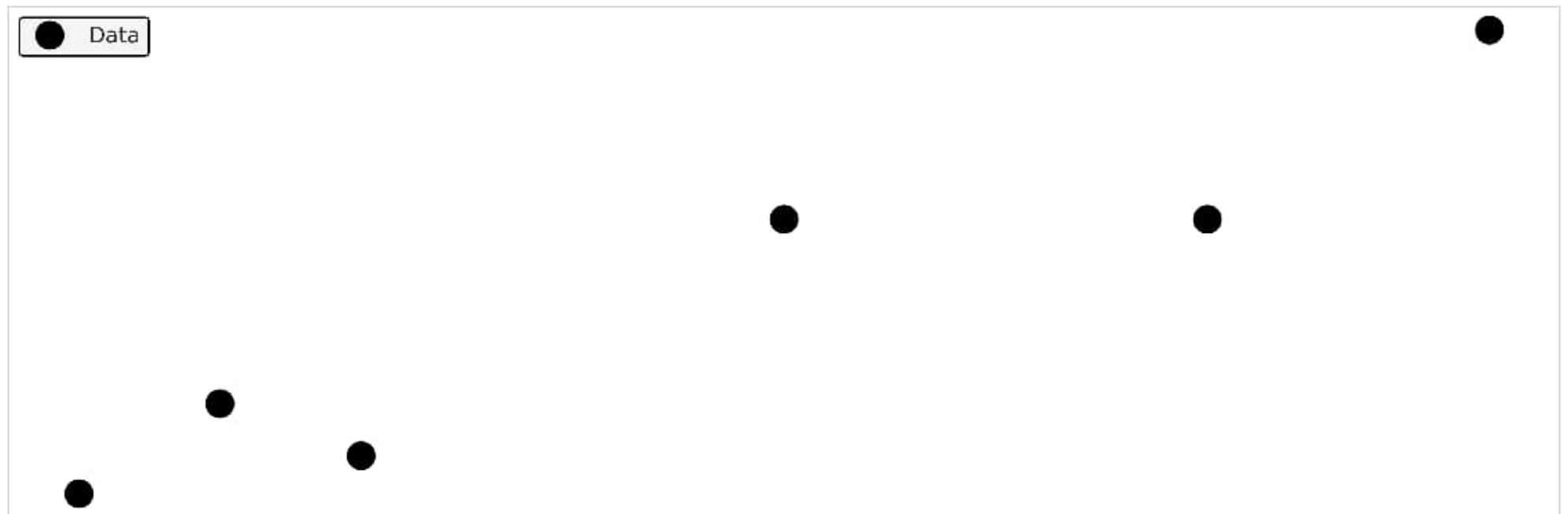
Occam's razor is about useful simplicity

- A model with more parameter can fit more complex data
 - Maybe fits real structure in the world
 - Maybe a model is just fitting noise
- When two models have the same fit
 - Prefer the simple one

Chatton's anti-razor

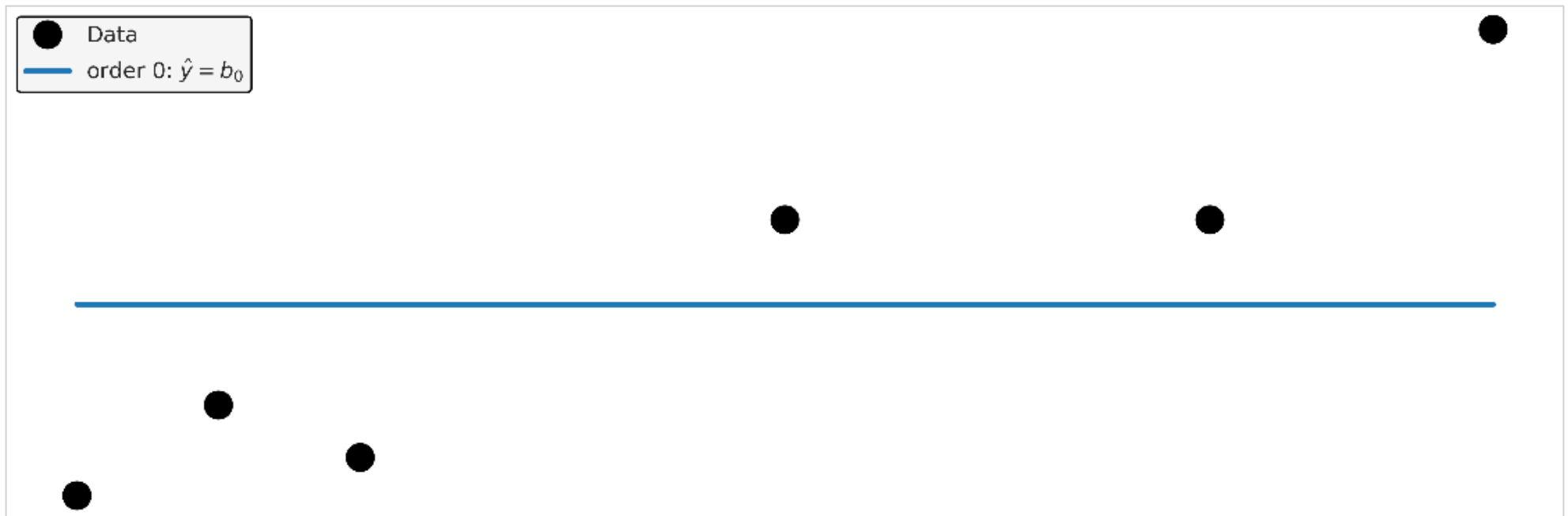
- Contemporary disputant of William of Ockham
- If three assumptions are not enough to explain the phenomena
 - A fourth will be necessary

We want to model some data



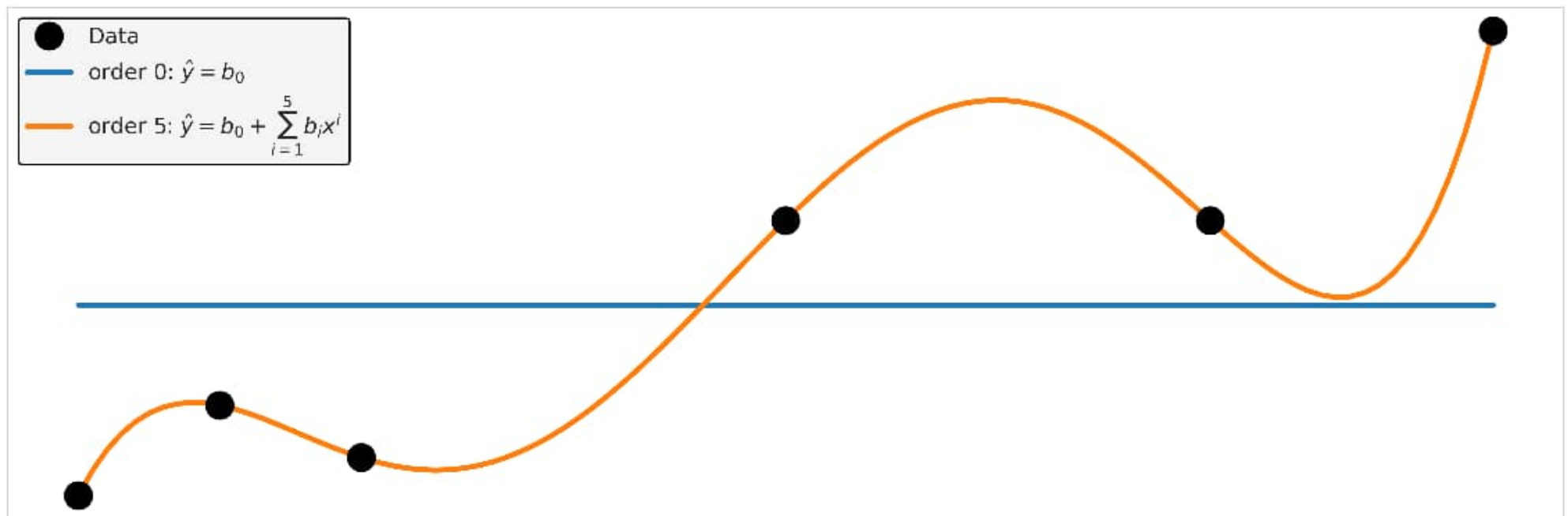
A model that underfits

- A 0th order model is one where the highest order term is x^0
 - Like a 0th order polynomial



A model that overfits

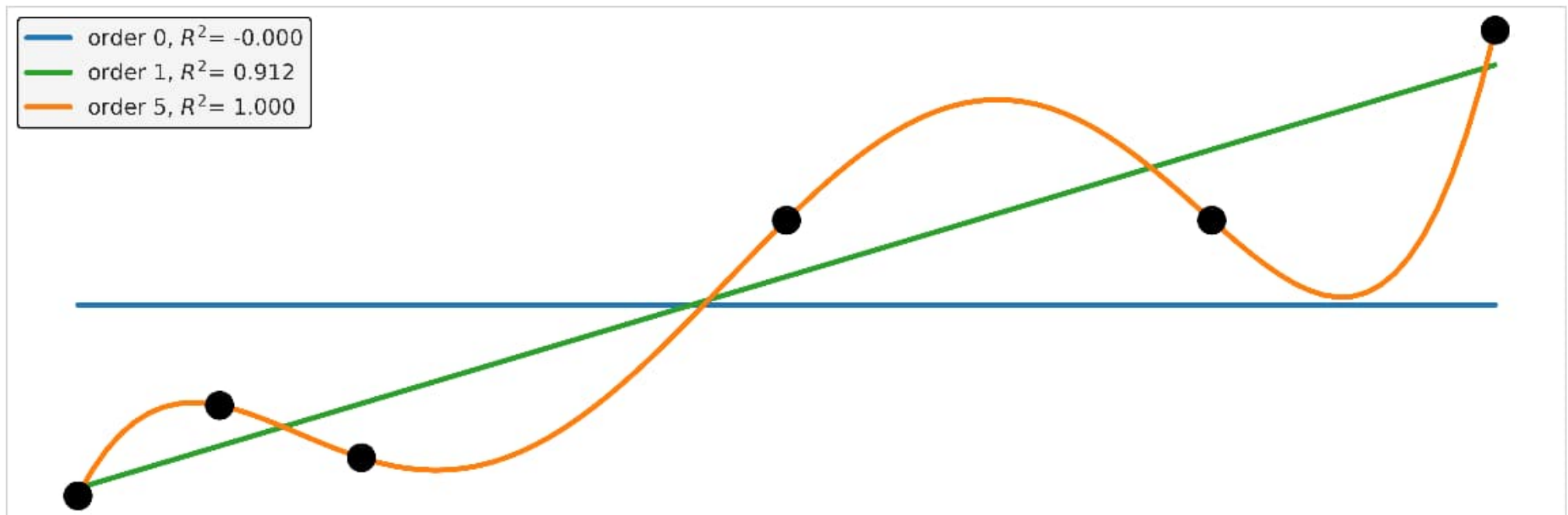
- A 5th order model is one where the highest order term is x^5
 - Like a 5th order polynomial



We want a model that balances overfitting and underfitting



R^2 increases with the number of parameters



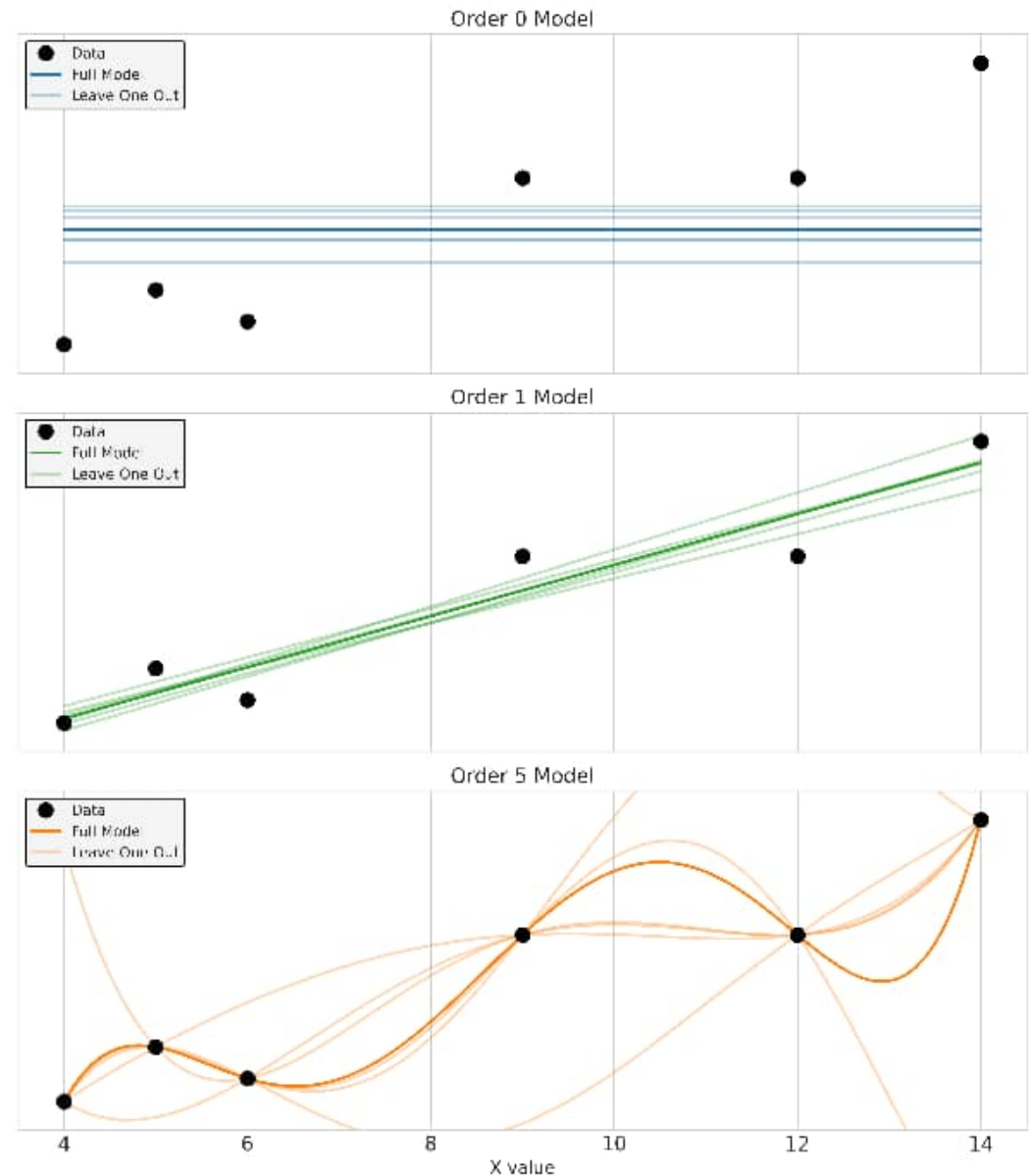
How well should a model fit?

- A model that is too simple is like a prior that is too strong
 - Choice of model determines results
 - The data doesn't speak
- A model that is too complex is like a prior that is too weak
 - Noise in the data overwhelms structure
- Model choice is just as important as prior choice
 - Like prior choice, it can be done right

Leaving out one data point

- Overfit model
 - Overly sensitive to data details
- Underfit model
 - Ignores important details
- In real model comparison
 - This is a key method

Polynomial Models with Leave-One-Out Cross-Validation



10.2 What makes a good model

Different scientific goals imply different comparison tools

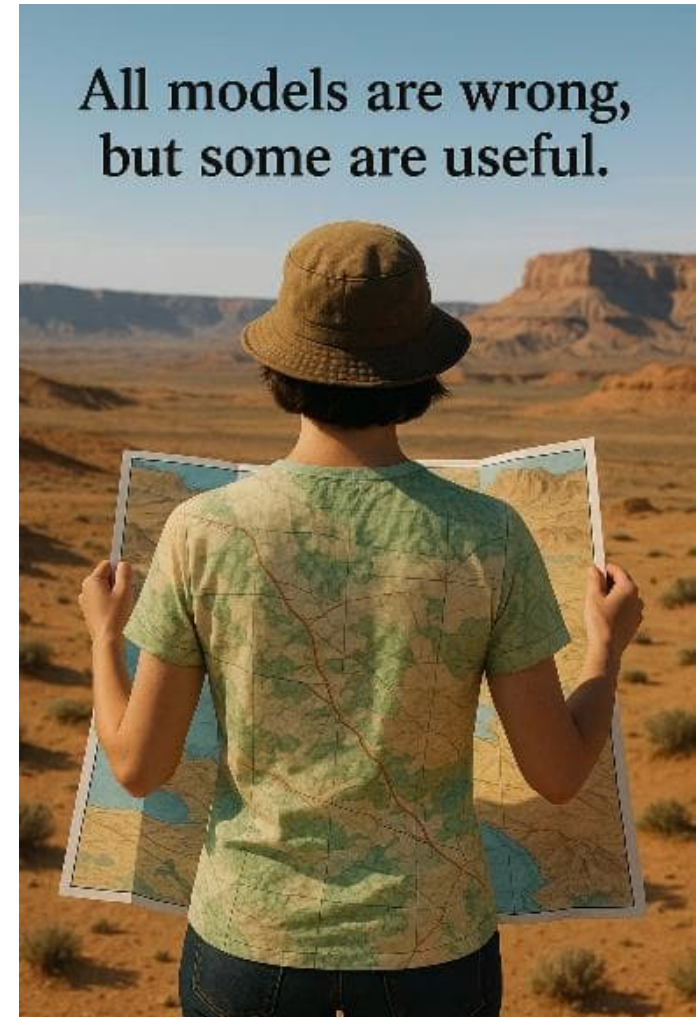
All models are wrong

- A perfect model would be as complex as reality
- Useful models simplify reality
- The question is not whether a model is true
- The question is whether it is useful for the goal

“All models are wrong, but some are useful.”

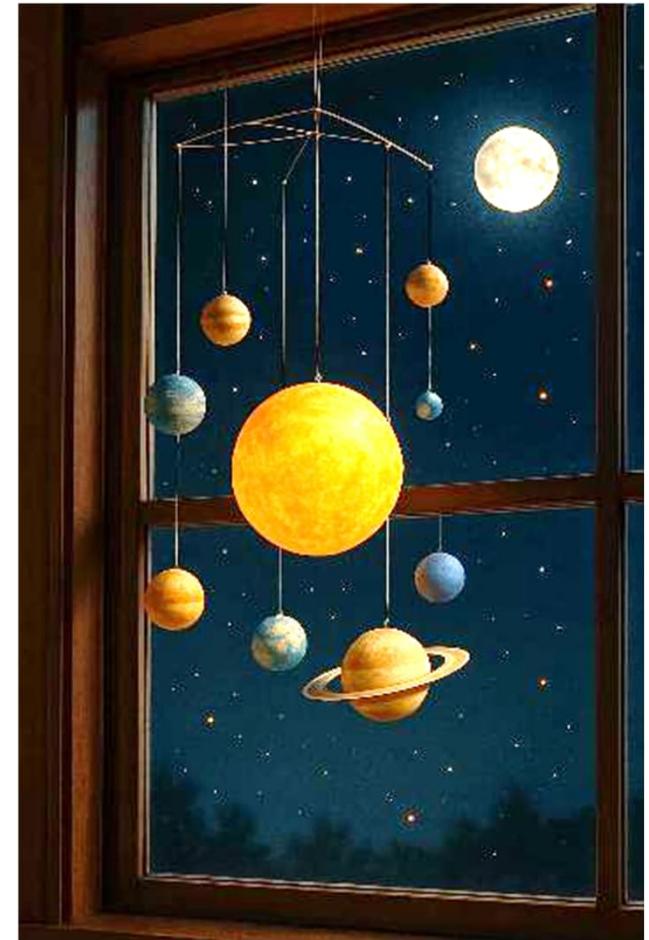
Some models are useful

- So what can they be useful for?
 - Hypothesis testing
 - Prediction
 - Theory building
- Different uses, different goals

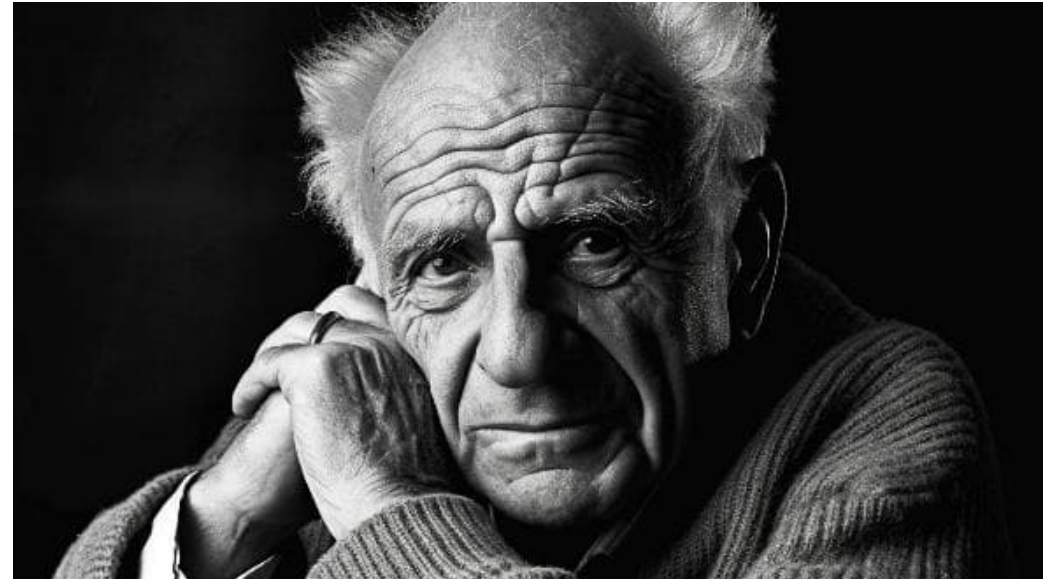


Theory building

- Models underlie our understanding
 - Especially scientific understanding
- Reflect important truths about reality
- Better models reflect important parts
 - More faithfully
- A good model must be:
 - Interpretable
 - Meaningful
 - Generalizable
- Science is: accepting good enough models



Hypothesis testing



- Karl Popper *The Logic of Scientific Discovery* 1934
 - Scientific theories can't be proven, but only falsified
- Practically this lead to:
 - Progress through falsifying simplistic theories
 - Statistical tests to reject simple theories
 - No different / no effect / no influence
- Science is: rejecting bad models

Prediction

- Up until the 1970s
 - Focus on finding 'true' parameter values
 - Focus on hypothesis testing
- 1970s: The link through information theory
 - Find the model 'closest' to the true model
 - Notice connection to predictive accuracy
- 1980s: The prediction revolution
 - Mervyn Stone 1983
 - A.P. Dawid 1984
 - Under the best model, future data is likely
- Science is: comparing candidate models

Model comparison depends on the goal

Goal	Good model means	Tools
Theory building	Interpretable and meaningful	posterior contrasts, effect sizes
Hypothesis testing	Rules out a simple model	ROPE/HDI, Bayes factor
Prediction	Predicts new data well	Tools of model comparison we will learn today

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When two models have nearly identical ability to predict new data, but model A has fewer parameters and model B..

1

Choose Model A, because equal prediction means the extra parameters in ...

2

Choose Model B, because a biophysical interpretation will probably mean ...

3

Choose Model A, because simpler models are generally closer to the true ...

4

Choose Model B, because a biophysical interpretation will probably mean ...

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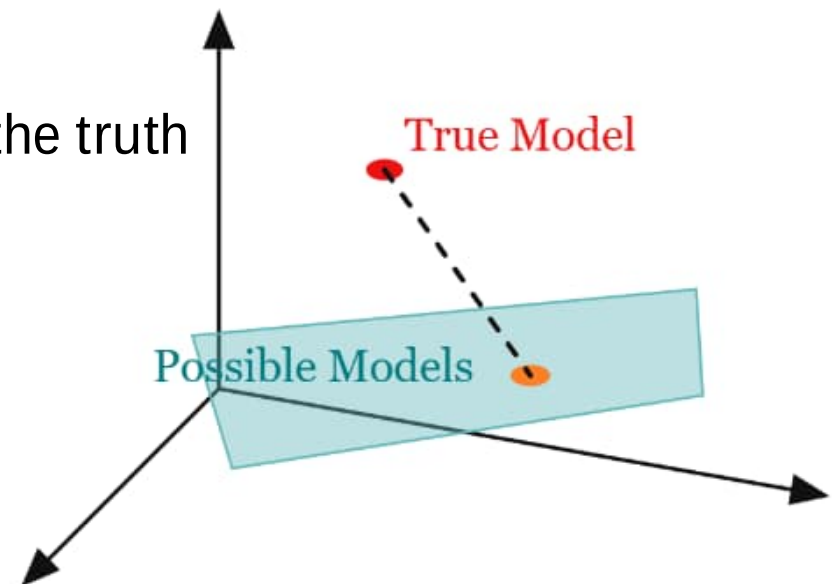
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10.3 Kullback-Liebler Divergence

Models that are closer to being the true model
predict better

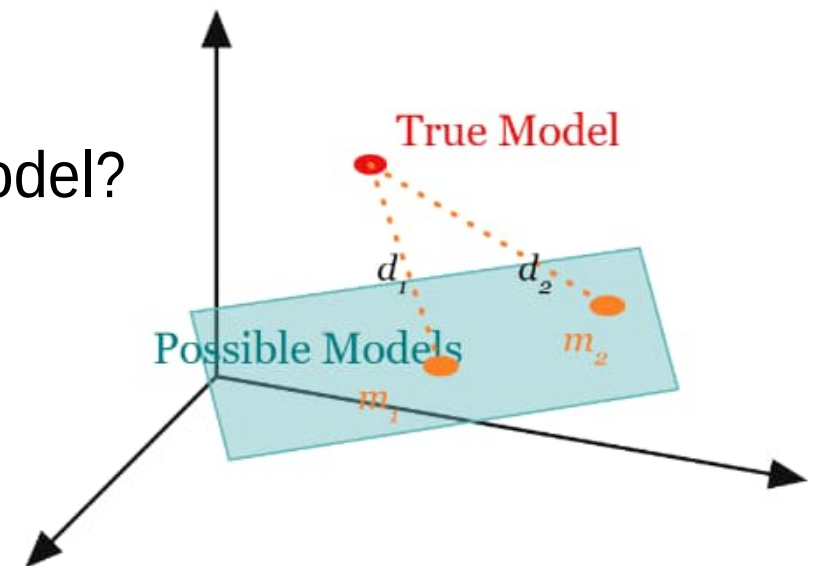
We do not have the true model

- Hypothesis testing assumes one model might be true
 - Either the null model is true or an alternative
- But we know that all models are false
 - Maybe the best model is the one closest to the truth



A distance measure might help

- A distance measure might
 - Give the distance from the true model
 - Allow comparison between models
- Can we measure distance from the true model?
 - By using new data
 - New data comes from the true model



A measure of distance between models

- The Kullback-Leibler divergence
 - The added “entropy” from using the wrong distribution
 - The cross entropy minus the entropy

$$D_{KL}(p, q) = \left(- \int_y dy p(y) \log q(y) \right) - \left(- \int_y dy p(y) \log p(y) \right) = H(p, q) - H(p)$$

$$\text{Entropy: } H(p) = - \langle \log p(y) \rangle_y = - \int_y dy p(y) \log p(y)$$

$$\text{Cross-entropy: } H(p, q) = - \langle \log q(y) \rangle_y = - \int_y dy p(y) \log q(y)$$

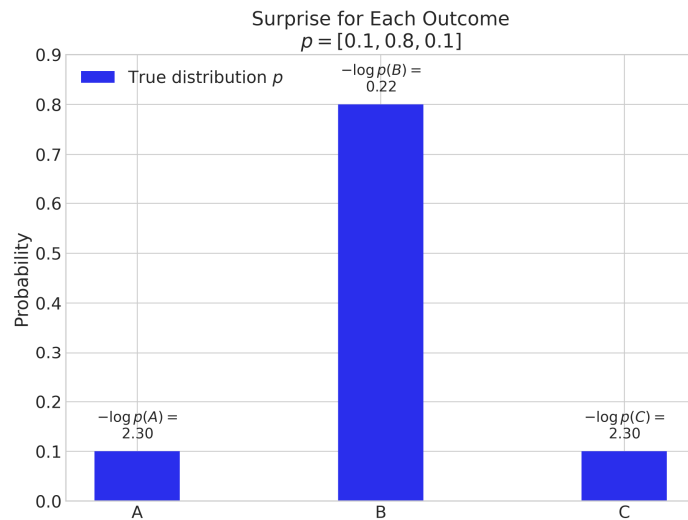
Surprise connects probability and prediction

$$H(p) = -\langle \log p(y) \rangle_y = -\int dy p(y) \log p(y)$$

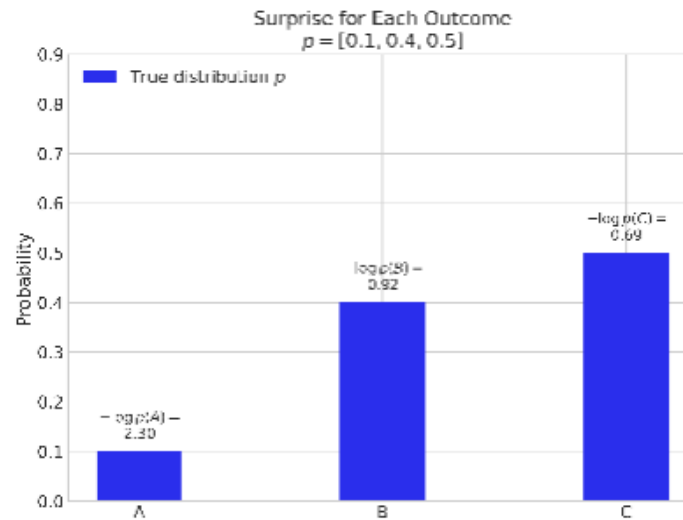
- Three concepts of entropy
 - The number of possible states
 - The uncertainty about the state
 - The information from learning the state
- For all three: the entropy of two separate systems should be additive

Entropy is a measure of how predictable reality is

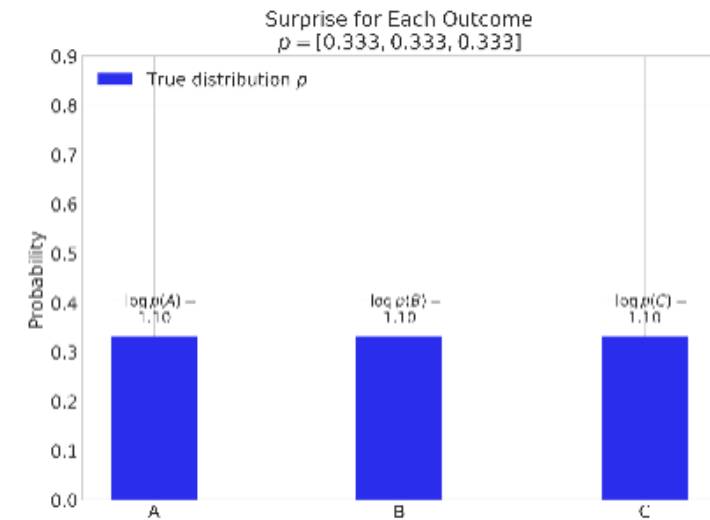
- If we make a guess according to probability
 - How likely are we to be correct?



$$H(p) = 0.1(2.30) + 0.8(0.22) + 0.1(2.30) = 0.64$$



$$H(p) = 0.1(2.30) + 0.4(0.92) + 0.5(0.69) = 0.943$$

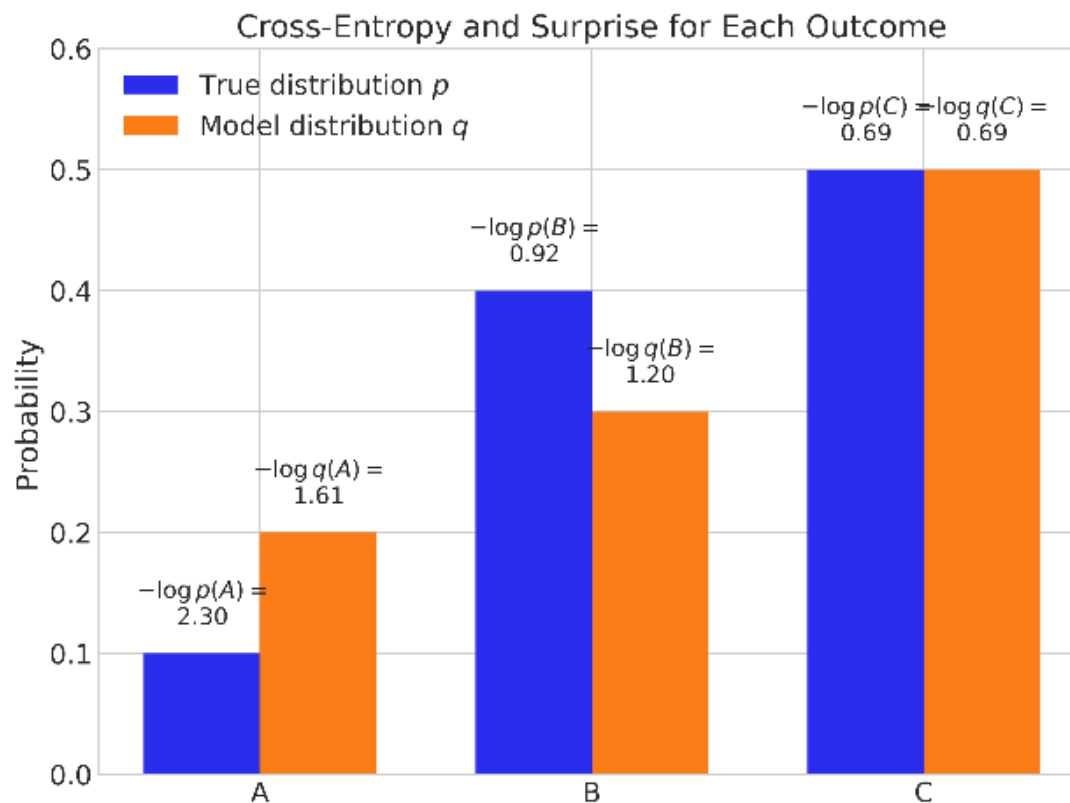


$$H(p) = 0.33(1.10) + 0.33(1.10) + 0.33(1.10) = 1.10$$

$$H(p) = -\langle \log p(y) \rangle_y = -\int_y dy p(y) \log p(y)$$

Cross-entropy is surprise for wrong guesses

- The cross-entropy is always larger than the entropy
 - Being wrong means being more surprised



$$H(p, q) = -\int dx p(x) \log q(x)$$

$$H(p) = 0.1(2.30) + 0.4(0.92) + 0.5(0.69) = 0.943$$

$$H(p, q) = 0.1(1.61) + 0.4(1.20) + 0.5(0.69) = 0.986$$

KL divergence measures extra surprise

- The difference between the cross entropy and the entropy
- The added uncertainty of using the wrong model
- An (asymmetric) measure of distance between two distributions

$$D_{KL} (p \| q) = H(p, q) - H(p)$$

Insert definitions

$$= \left(- \int dp \, p(x) \log q(x) \right) - \left(- \int dp \, p(x) \log p(x) \right)$$

Rearrange terms

$$= \int dp \, p(x) \log p(x) - \int dp \, p(x) \log q(x)$$

Combine integrals

$$= \int dp \, p(x) (\log p(x) - \log q(x))$$

The average (over the true distribution) of how wrong we are (in units of surprise)

In our case: the model from the truth

- The distance of the model's distribution from the true data distribution
 - The true likelihood vs the posterior predictive likelihood

$$D_{KL} (p \| q) = \int dx \, p(x) (\log p(x) - \log q(x))$$

$$D_{KL} (p(y) \| p(y|m)) = \int_y dy \, p(y) (\log p(y) - \log p(y|m))$$

Here y means the probability density of data, not the specific data we collected.

The score

- $\int dy p(y)p(y|m)$ is the average likelihood of data under the model
 - Averaged over the true distribution of data
- We could estimate this by collecting new data and calculating its likelihood under the model

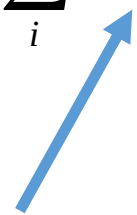
$$D_{KL} \left(p \parallel p(y|m) \right) = \int_y dy p(y) (\log p(y) - \log p(y|m))$$

$$= - \int_y dy p(y) \log p(y|m) + H(p)$$

$$\approx - \sum_i \log p(\tilde{y}_i | m) + H(p)$$

$\tilde{y}_i$ means an i -th actual new data point

The score



Calculating the score

- To find likelihood of new data under the model
 - Average over the posterior distribution under the model

$$D_{KL} \left(p \parallel p(y | m) \right) \approx -\sum_i \log p(\tilde{y}_i | m) + H(p)$$

$$\sum_i \log p(\tilde{y}_i | m) = \sum_i \int_{\theta} d\theta \log p(\tilde{y}_i | \theta, m) p(\theta | m)$$

Often people calculate deviance instead of score
Different units, same idea

$$\text{deviance} = -2 \cdot \text{Score} = -2 \sum_i \log p(\tilde{y}_i | m)$$

The difference of two D_{KL} for two models

- Is the difference of their scores!

$$\begin{aligned} D_{KL} \left(p \parallel p(y | m_1) \right) &- D_{KL} \left(p \parallel p(y | m_2) \right) \\ &= \left(-\sum_i \log p(\tilde{y}_i | m_1) + H(p) \right) - \left(-\sum_i \log p(\tilde{y}_i | m_2) + H(p) \right) \\ &= \left(-\sum_i \log p(\tilde{y}_i | m_1) \right) - \left(-\sum_i \log p(\tilde{y}_i | m_2) \right) \end{aligned}$$

To compare models we calculate the score

- Each score is meaningless, only the comparison matters
- Cannot be actually calculated on existing data

$$D_{KL} \left(p \parallel p(y | m) \right) \approx - \sum_i \log p(\tilde{y}_i | m) + H(p)$$

$$D_{KL} \left(p \parallel p(y | m_1) \right) - D_{KL} \left(p \parallel p(y | m_2) \right) = \left(- \sum_i \log p(\tilde{y}_i | m_1) \right) - \left(- \sum_i \log p(\tilde{y}_i | m_2) \right)$$

Intuitively: how wrong a model is about what the real world is like (the KL divergence from reality) can be characterized by the likelihood it associates with new data.

Up to a constant factor

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Suppose you know the posterior distribution of the parameters and you can use it to calculate the posterior...



- ① The prior predictive distribution
- ② The posterior distribution of the parameters
- ③ The distribution which generates future observations
- ④ The sampling distribution of the estimator



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10.4 Information criteria

Methods for approximating the score without getting new data

Information criteria estimate out-of-sample prediction

- We want expected predictive accuracy on new data
- We only have the observed data
 - Our predictive accuracy is always better on observed data than new data
- Information criteria estimate the optimism from fitting and testing on the same data
- Different criteria estimate this correction differently

$$\sum_i \log p(\tilde{y}_i | m) = \sum_i \log \int d\theta p(\tilde{y}_i | \theta, m) p(\theta | m)$$

One approximation is that our optimism is measured by how quickly the log likelihood changes with θ

$$\approx \sum_i \log \int d\theta p(y_i | \theta, m) p(\theta | m) - \sum_i \text{Var}_\theta (\log p(y_i | \theta, m))$$

Several criteria estimate predictive accuracy

Criterion	In sample log density	Optimism term
AIC	$-2 \sum_i \log p(y_i \hat{\theta}_{MLE}, m)$	$2k$ where k is the number of model parameters
DIC	$-2 \sum_i \log p(y_i \bar{\theta}, m)$	$2 \left[\sum_i \log p(y_i \bar{\theta}, m) - \mathbb{E}_{\theta y} \sum_i \log p(y_i \theta, m) \right]$
WAIC	$-2 \sum_i \log \mathbb{E}_{\theta y} [p(y_i \theta, m)]$	$2 \sum_i \text{Var}_{\theta y} [\log p(y_i \theta, m)]$

AIC is the Frequentist information criterion

$$AIC = -2\log p(y|\hat{\theta}, m) + 2k$$

- $-2\log p(y|\theta, m)$: Smaller when model has good fit to observed data
- $2k$: Smaller for simpler models
 - k is the number of parameters
- Lower AIC is better

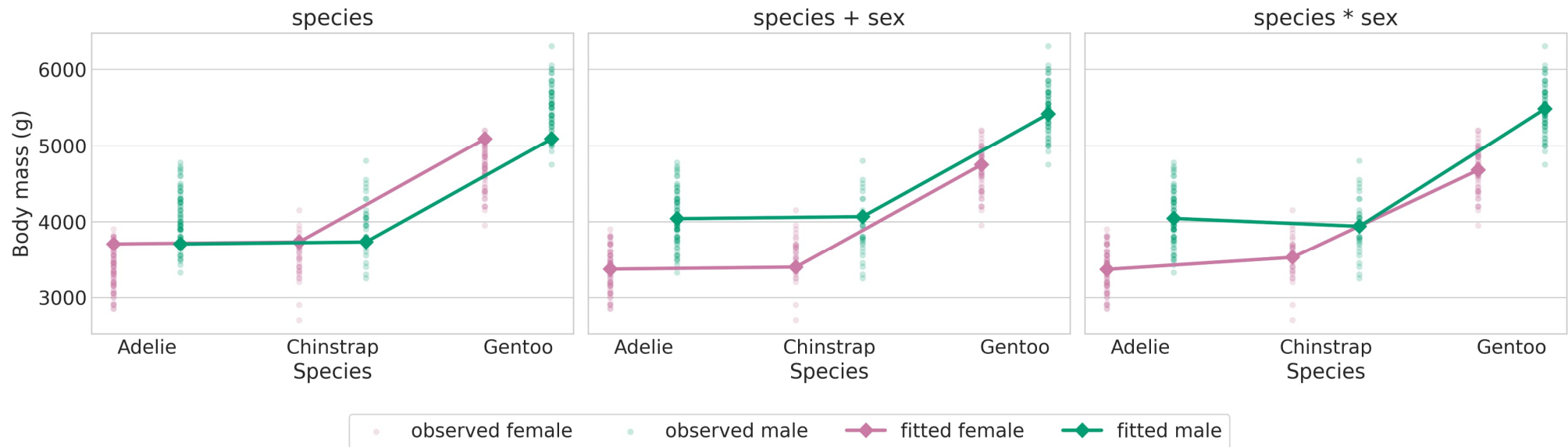
AIC has limitations

- Uses a point estimate of in sample likelihood
 - Neglects uncertainty
- Uses parameter counting
 - The number of parameters measures model complexity
 - Proportional to variance for linear regression models
 - But assumes:
 - Parameters are not correlated
 - Parameters are all regression coefficients
 - Model is not hierarchical or otherwise structures

Comparing different models

- Fit 3 models to the penguin bodyweight data

Observed penguin body mass and fitted means



AIC in an applied model comparison

- Compare candidate regression or ANOVA models
- Smaller AIC means better estimated predictive performance
- Differences matter more than absolute values

```
smf.ols(formula, data=penguins_model_df).fit()  
ols_fit.aic
```

	model	formula	parameters	AIC	delta_AIC
2	species * sex	body_mass_g ~ species * sex	6	4770.2	0.0
1	species + sex	body_mass_g ~ species + sex	4	4783.6	13.4
0	species	body_mass_g ~ species	3	5032.5	262.3

WAIC is a Bayesian information criterion

- Uses the full posterior to estimate the score
 - Accounts for parameter uncertainty for in sample likelihood
 - Uses the shape of the posterior directly to estimate optimism

$$\sum_i \log p(\tilde{y}_i | m) = \sum_i \int_{\theta} d\theta \log p(y_i | \theta, m) p(\theta | m) - \sum_i \text{Var}_{\theta}(\log p(y_i | \theta, m))$$

$$\text{WAIC} = -2 \sum_i \int_{\theta} d\theta \log p(y_i | \theta, m) p(\theta | m) + 2 \sum_i \text{Var}_{\theta}(\log p(y_i | \theta, m))$$

WAIC in an applied model comparison

- WAIC table gives estimate uncertainty
 - 'Best' model may not be much better
- WAIC model also shows test for influential data points
 - If one data point is overly influential, WAIC is biased estimate of score

```
az.compare(penguin_idata, ic="waic")
```

	rank	elpd_waic	p_waic	elpd_diff	weight	se	dse	warning	scale
species * sex	0	-2386.290200	6.967973	0.000000	0.862056	12.175076	0.000000	False	log
species + sex	1	-2392.875092	4.960643	6.584892	0.137944	12.005834	4.269239	False	log
species	2	-2517.111104	3.607536	130.820904	0.000000	11.230931	10.687399	False	log

- p_waic estimates the model complexity (the number of effective parameters)
- weight can be used to get improved predictions by using a weighted combination of models
- dse is the standard error of the difference
- Warning means that there are influential points

Predictive power is not the only criterion

- Three different criteria for a good model
 - Building theories
 - Testing hypotheses
 - Predicting the future
- When evaluating models, we should consider all three:
 - Does it have predictive power?
 - Does it inform us about our scientific hypotheses?
 - Does it give us information about meaningful parameters?
- In different situations, these may have different importance?

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Suppose Model A and Model B have identical WAIC values.
They are both simple regression models, but model B has...



- ① Model A has better predictive value
- ② Model A has worse predictive value
- ③ Model B has lower in sample log predictive density
- ④ Model B has higher in sample log predictive density



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10.5 Cross-validation

Testing prediction by fitting to part of the data and testing on the other part

WAIC is only an estimate of true validation

- WAIC estimates the ability to predict new data
- Testing the model on actual new data is called validation

Validation and cross-validation

- Strategies for validation

- True validation
 - Go out and collect a new data set
- Withheld data
 - Divide data into train and test
 - Sometimes into train, test, and validate
- Cross-validation
 - Train on subsets of data
 - Test on others
 - Repeat over subsets

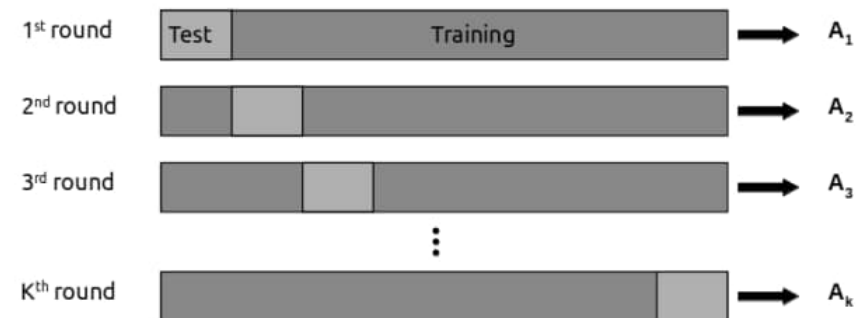
True validation is expensive and time consuming

Withheld data is vulnerable to leakage
There is a bias variance tradeoff

Cross-validation requires multiple model fits

K-fold cross-validation

- Divide the data into k chunks
 - For each chunk
 - Fit model to the rest of the chunks
 - Evaluate log likelihood on that chunk
 - Average predictive ability across chunks
-
- Widely used in machine learning
 - Problematic in Bayesian analysis
 - Because fitting models is expensive



Leave-one-out cross-validation

- Make N chunks of size 1
- Approximate instead of refitting the model

Expected log predictive density

$$\text{ELPD} = \int d\tilde{y} \, p(\tilde{y}) \log \int d\theta \, p(\tilde{y} | \theta) p(\theta | y)$$

$$\text{ELPD}_{\text{LOO-CV}} = \sum_i \log \int d\theta \, p(y_i | \theta) p(\theta | y_{-i})$$

LOO-CV is an unbiased estimate of ELPD

- Where WAIC corrects a biased estimate

$$\text{ELPD} = \int d\tilde{y} p(\tilde{y}) \log \int d\theta p(\tilde{y} | \theta) p(\theta | y)$$

$$\text{ELPD}_{\text{WAIC}} = \sum_i \log \int d\theta p(y_i | \theta) p(\theta | y) + \text{optimism term}$$

$$\text{ELPD}_{\text{LOO-CV}} = \sum_i \log \int d\theta p(y_i | \theta) p(\theta | y_{-i})$$

LOO is expensive if done literally

- We are taking the integral over the wrong probability
- This trick is called importance sampling

$$\begin{aligned}\text{ELPD}_{\text{LOO-CV}} &= \sum_i \log \int d\theta p(y_i | \theta) p(\theta | y_{-i}) \\ &= \sum_i \log \int d\theta p(y_i | \theta) \frac{p(\theta | y_{-i})}{p(\theta | y)} p(\theta | y) \\ &= \sum_i \log \frac{1}{S} \sum_s p(y_i | \theta_s) \frac{p(\theta_s | y_{-i})}{p(\theta_s | y)} \\ &= \sum_i \log \frac{1}{S} \sum_s w_s p(y_i | \theta_s)\end{aligned}$$

If we had

$$\int d\theta p(y_i | \theta) p(\theta | y)$$

We would write

$$\int d\theta p(y_i | \theta) p(\theta | y) \approx \frac{1}{S} \sum_s p(y_i | \theta_s)$$

Now we have

$$\int d\theta f(\theta) p(\theta | y)$$

So we can write

$$\int d\theta f(\theta) p(\theta | y) \approx \frac{1}{S} \sum_s f(\theta_s)$$

Importance sampling approximates LOO

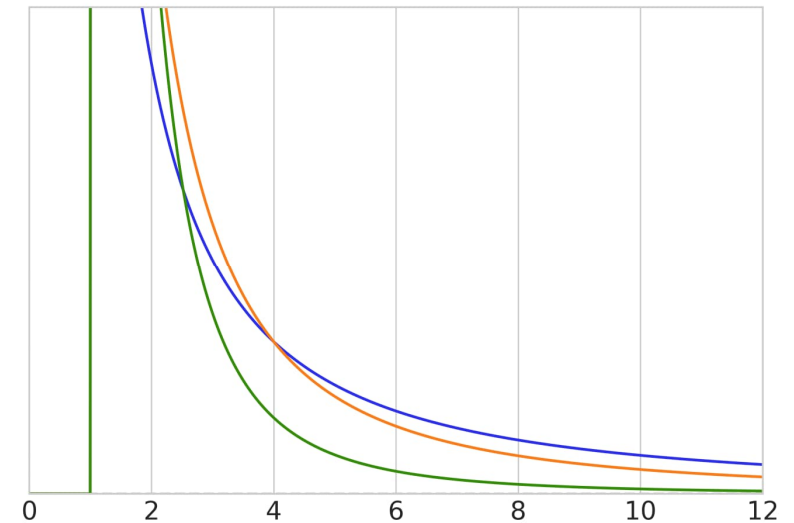
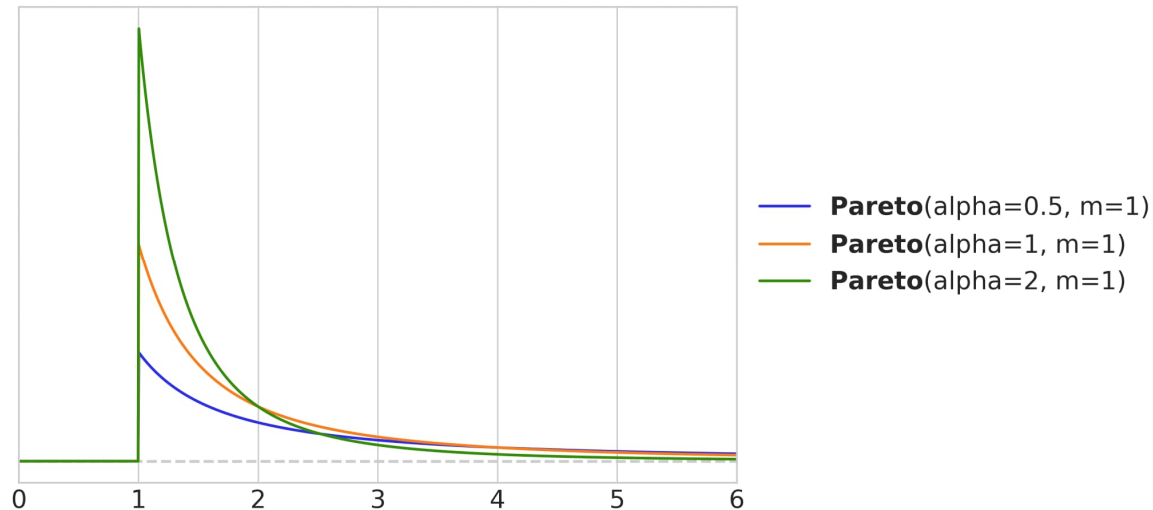
- This approximates the expected log likelihood of new data

$$\text{ELPD}_{\text{LOO-CV}} = \sum_i \log \frac{1}{S} \frac{\sum_s w_s p(y_i | \theta_s)}{\sum_s w_s}$$

$$w_s = \frac{1}{p(y_i | \theta_s)}$$

PSIS stabilizes the approximation

- The Pareto distribution has wider tails
 - Like the Student's t distribution
 - But much wider
- Replace extreme weights (usually top 20%)
 - Using Pareto distribution fit to the rest of the weights
 - Unreliable if shape parameter > 1 !



Pareto k is a diagnostic

- In importance sampling
 - You sample the proposal distribution instead of the target distribution
 - And then weight the samples according to the ratio
- This has a problem if the importance weight is too large

Target distribution: $p(\theta | y_{-i})$

Proposal distribution: $p(\theta | y)$

Importance weight: $\frac{p(\theta | y_{-i})}{p(\theta | y)} \propto \frac{1}{p(y_i | \theta)}$

PSIS-LOO is usually our preferred comparison tool

- Pareto-smoothed importance sampled leave one out cross validation
- Estimates the predicted likelihood of new data
 - Can be used to compare the predictive power of models
- Need to check diagnostic: the shape parameter
 - $k \geq 1$ suggests LOO-CV estimates may be unreliable
 - In this case, full cross-validation is required

Comparing model comparison criteria

	model	formula	parameters	AIC	delta_AIC
2	species * sex	body_mass_g ~ species * sex	6	4770.2	0.0
1	species + sex	body_mass_g ~ species + sex	4	4783.6	13.4
0	species	body_mass_g ~ species	3	5032.5	262.3

`az.compare(penguin_idata, ic="loo")`

	rank	elpd_waic	p_waic	elpd_diff	weight	se	dse	warning	scale
species * sex	0	-2386.290200	6.967973	0.000000	0.862056	12.175076	0.000000	False	log
species + sex	1	-2392.875092	4.960643	6.584892	0.137944	12.005834	4.269239	False	log
species	2	-2517.111104	3.607536	130.820904	0.000000	11.230931	10.687399	False	log

	rank	elpd_loo	p_loo	elpd_diff	weight	se	dse	warning	scale
species * sex	0	-2386.303741	6.981514	0.000000	0.861795	12.176171	0.000000	False	log
species + sex	1	-2392.884856	4.970406	6.581114	0.138205	12.006771	4.269849	False	log
species	2	-2517.116369	3.612801	130.812628	0.000000	11.231261	10.688258	False	log

Important caveats about predictive power measures

- You cannot use them on models with different likelihoods
 - Changing the likelihood is like changing the scale of measurement
- They apply to hierarchical models but not trivially
 - Are you predicting:
 - New observations with the same groups
 - New observations with different groups
 - New groups and not new observations
- Influential observations or a large Pareto smoothing constant should not be ignored
- More than one model may be viable
 - Sometimes differences are small
 - Sometimes a model can be preferred despite lower predictive power

< Select another question

Edit question

Go to wooclap.com and use the code **ECFXCHN**



A model has excellent R^2 but poor PSIS-
LOO. The most likely explanation is:



- ① Underfitting
- ② Overfitting
- ③ Poor prior predictive performance
- ④ Poor posterior predictive performance



Click on the projected screen to start the question

wooclap



100%



10.6 Updated Bayesian workflow

Model comparison is one step in a larger analysis

Model comparison belongs inside the workflow

- 1 Scientific question
- 2 Candidate models
- 3 Prior predictive check
- 4 Sample
- 5 Diagnostics
- 6 Posterior predictive check
- 7 Compare predictive accuracy
- 8 Interpret model differences

Watch for

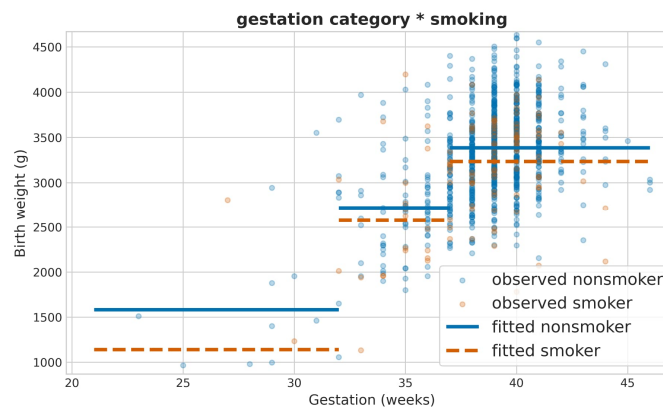
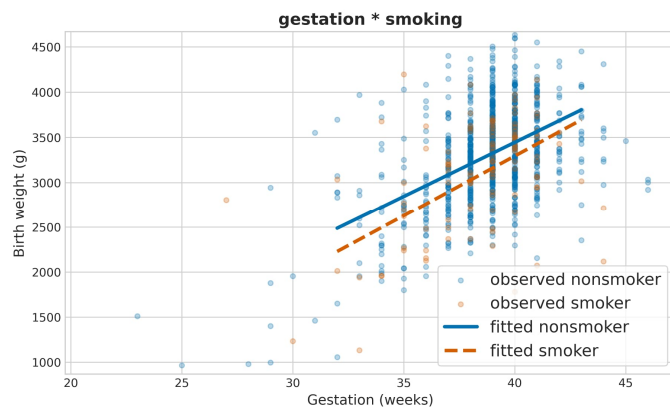
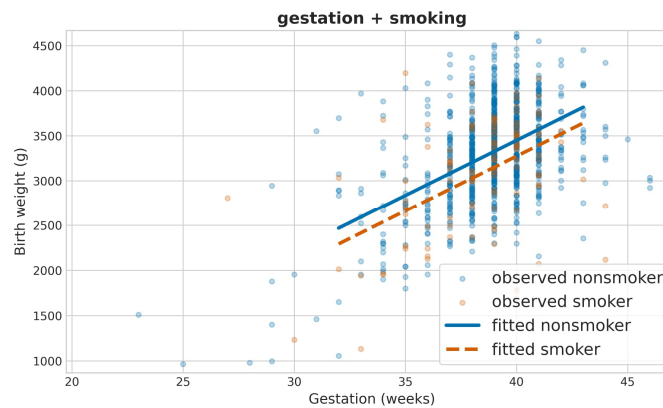
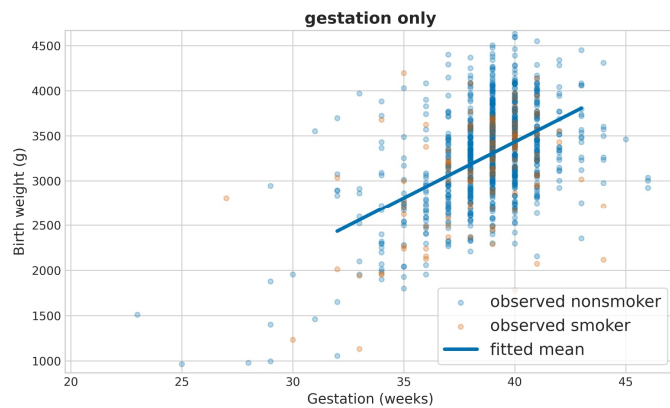
- Compare models that answer the scientific question
- Never compare models before checking diagnostics
- Predictive ranking usually doesn't replace interpretation

Effect size and prediction answer different questions

Quantity	Question
R^2	How much variance does the model explain?
partial R^2 , η^2	What does one term add?
Cohen's f	Effect size in residual-variance units
AIC / WAIC / LOO	Which model predicts new data better?

- Effect size is about interpretation
- Predictive comparison is about expected future performance
- A model can have a meaningful effect but weak predictive advantage

Returning to birth weight and gestation age



- Looking at this with our new lens
 - What were the effects
 - What were the effect sizes
 - Which model has the best predictive power
 - Which model deepens our understanding

Effect sizes for the models and the factors

- Effect sizes of smoking in the interaction model are largest

	R2	partial_R2_gestation	partial_R2_smoking	partial_R2_gestation_smoking
model				
gestation only	[0.254, 0.339]	[0.254, 0.339]		
gestation + smoking	[0.262, 0.344]	[0.248, 0.330]	[0.002, 0.027]	
gestation * smoking	[0.262, 0.347]	[0.234, 0.329]	[0.000, 0.529]	[0.000, 0.469]
gestation category + smoking	[0.259, 0.344]	[0.250, 0.334]	[0.010, 0.051]	
gestation category * smoking	[0.261, 0.345]	[0.255, 0.340]	[0.011, 0.051]	[0.005, 0.014]

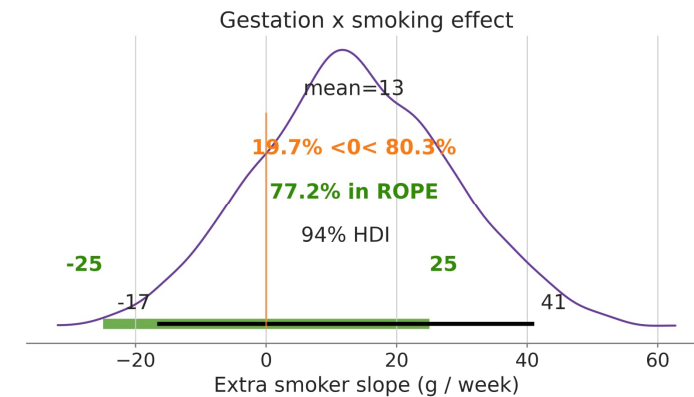
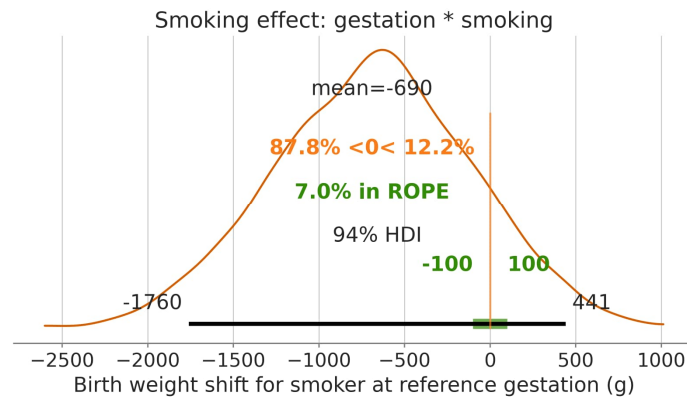
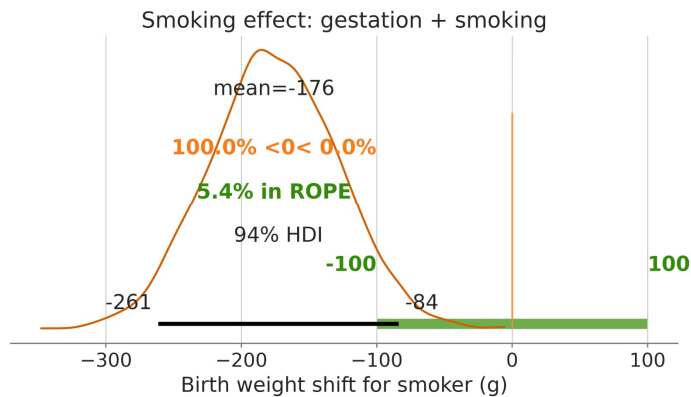
Model comparison

- The standard error of all the differences is on the scale of the differences themselves

	rank	elpd_loo	p_loo	elpd_diff	weight	se	dse	warning	scale
gestation + smoking	0	-7471.824102	5.291485	0.000000	2.765475e-01	23.594702	0.000000	False	log
gestation * smoking	1	-7473.553619	8.081545	1.729517	1.653462e-16	23.931622	1.552772	False	log
gestation category + smoking	2	-7475.223233	8.458199	3.399131	4.838473e-01	25.369752	14.967250	False	log
gestation only	3	-7477.413670	4.484762	5.589568	2.396051e-01	24.116911	3.943605	False	log
gestation category * smoking	4	-7478.489826	13.440635	6.665724	0.000000e+00	25.801391	15.196587	False	log

The central parameter of interest is the effect of smoking

- A small effect in both leading models



Summarizing the smoking / birthweight analysis

- For the model as a whole
 - All models have similar overall effect size
 - All models have similar predictive power
- For the parameter of interest
 - Simple additive model gives tightest bounds
 - Interaction model has largest effect size but large uncertainty
- Taken together
 - Across models there is a small but consistent effect of smoking
 - Across models there is no strong evidence for an interaction

10.7 Updated Bayesian workflow

Model comparison is one step in a larger analysis

Next time: complex data one last time!

- Combining regression with hierarchical models!

